

A-CASA Dataset Documentation

Data Card, File Manifest, and Verifiable References

Version 1.0 — April 2026

Contents

1	Dataset Overview	2
1.1	Provenance statement	2
1.2	Dataset summary	2
1.3	Stratum and tier distribution	2
2	Synthetic Data Generation Methodology	3
3	File Manifest	3
3.1	Directory structure	3
3.2	Column schema (flat format)	4
3.3	SHA-256 checksums	4
4	Lethality Scoring Table — Data Sources	4
5	NER Entity Class Definitions — Reference Basis	4
6	Verifiable References for All Empirical Claims	4
6.1	Korean suicide epidemiology	4
6.2	C-CASA and C-SSRS instruments	5
6.3	Pre-hospital EMS data quality	6
6.4	NLP models and toolkits	6
6.5	Blockchain and infrastructure	6
6.6	Dispatch triage NLP baselines	7
6.7	Supplementary lethality and toxicology references	7
6.8	Regulatory and data governance references	7
7	How to Reproduce the Experiments	8
8	Limitations and Ethical Considerations	8

1 Dataset Overview

1.1 Provenance statement

The A-CASA paper (“A-CASA: A Clinically-Grounded NLP Triage Framework with Blockchain Data Assurance for Suicide-Risk Stratification in Unstructured Pre-Hospital 119 Emergency Dispatch Logs”) reports results on **4,218 retrospective 119 emergency dispatch records** collected from a metropolitan EMS dispatch centre in Seoul, Republic of Korea (January 2022–December 2024).

Access to the real dataset requires:

1. A signed data-sharing agreement with the institutions participating into the EMS dispatch centre.
2. IRB approval from the requesting institution.
3. Compliance with the Korean Personal Information Protection Act (*Gae-in-jeong-bo Bo-ho-beop*, <https://www.pipc.go.kr>).

To enable full reproducibility of the experimental pipeline, we provide a **synthetic proxy dataset** generated to match the real cohort’s statistical properties. The synthetic data replicates: stratum distribution, call duration distribution, informant type proportions, and audio feature distributions. Synthesis methodology is described in Section 2.

1.2 Dataset summary

Table 1: A-CASA PoC dataset summary statistics.

Attribute	Value
Total records	4,218
Train / Val / Test split	2,530 / 844 / 844 (60/20/20, stratified)
Stratification variable	C-CASA stratum + acuity level
Random seed	42
Label noise rate	6% (simulates real-world annotation variability)
Time period (synthetic)	January 2022 – December 2024
Location (synthetic)	Seoul metropolitan districts (6 gu)
Text language	Korean (romanised exemplars)
License	CC BY 4.0 (synthetic data only)

1.3 Stratum and tier distribution

Table 2: Stratum and risk-tier distribution (synthetic dataset, $n = 4,218$).

C-CASA stratum	Risk tier	n	%
EXPLICIT_INTENT	IMMINENT	890	21.1
INFERRED_INTENT	HIGH	1209	28.7
UNKNOWN_INTENT	MODERATE	1010	23.9
NO_INTENT	LOW	633	15.0
NOT_ENOUGH_INFO	LOW/Review	476	11.3
Total		4,218	100

2 Synthetic Data Generation Methodology

The synthetic dataset was generated using `data_generator.py` (Supplementary File 2) with the following design principles:

1. **Stratum-conditional text templates.** Each record is drawn from a stratum-specific template pool encoding the characteristic lexical, syntactic, and informant patterns of each C-CASA category. Templates use romanised Korean dispatch expressions derived from published dispatch protocol documentation [6, 7].
2. **Lethality-conditioned entity sampling.** Self-harm method entities are sampled conditioned on the target stratum, with lethality scores drawn from Table 4 (calibrated on KDCA statistics [2]).
3. **Audio feature simulation.** Prosodic features (speech rate, pause ratio, affect score) are drawn from stratum-conditioned Gaussian distributions whose parameters match published crisis-call acoustic feature ranges [22, 23].
4. **Label noise injection.** 6% of records have their stratum label randomly flipped to a different stratum, simulating the real-world inter-annotator disagreement rate reported in the paper ($\kappa = 0.81$, corresponding to approximately 8% pairwise label disagreement for a 5-class problem) [26].
5. **Temporal distribution.** Timestamps are uniformly drawn from the 36-month study window (January 2022 – December 2024), matching the real cohort’s collection period.

3 File Manifest

3.1 Directory structure

acasa-poc/	
data/	
acasa_dispatch.jsonl	# Full dataset (nested
JSON, 4,218 records)	
acasa_dispatch.csv	# Full dataset (original
nested format)	
acasa_dispatch_flat.jsonl	# Full dataset (flat, all
columns)	
acasa_dispatch_flat.csv	# Full dataset (flat, CSV)
checksums.sha256	# SHA-256 checksums for
all files	
splits/	
acasa_train.jsonl	# Training split (n=2,530)
acasa_train.csv	
acasa_val.jsonl	# Validation split (n=844)
acasa_val.csv	
acasa_test.jsonl	# Test split (n=844)
acasa_test.csv	
data_generator.py	# Synthetic data generator
nlp_pipeline.py	# NLP pipeline (5 modules)
casa_fsm.py	# C-CASA FSM (47 rules)
ml_ensemble.py	# Transformer proxy +
ensemble	
blockchain_layer.py	# Hyperledger Fabric
simulation	
run_experiment.py	# Full experiment runner

3.2 Column schema (flat format)

Column	Type	Description
id	string	Record identifier (ACASA-00001 to ACASA-04218)
raw_text	string	Korean 119 dispatch log (romanised + key Korean terms)
timestamp	string	ISO 8601 call timestamp (synthetic, 2022–2024 window)
location	string	Seoul district (<i>gu</i>) where call originated
duration_sec	float	Call duration in seconds [30, 479]
informant_type	string	Caller identity: self , family , bystander , multiple
speech_rate	float	Simulated syllables/second (ASR feature)
pause_ratio	float	Simulated pause proportion [0, 1]
affect_score	float	Simulated emotional affect score [0, 1]
asr_confidence	float	Simulated ASR recognition confidence [0, 1]
true_stratum	string	Ground-truth C-CASA stratum (5 categories)
risk_tier	string	Ground-truth A-CASA risk tier (4 categories)

3.3 SHA-256 checksums

```

a91be9a436131584b13103dfb41b40ac...  acasa_val.csv
9fbdc24cd5313e57e9a607e93707e189...  acasa_test.csv
04527650e911d11ad9a9f6214802708d...  acasa_train.csv
fe288415a84aef759b61fc72ad32078f...  acasa_val.jsonl
94efb6e9192f358c0685ccf2e0661d0a...  acasa_train.jsonl
9f6f19b1caace8b611296d3fa98888d7...  acasa_test.jsonl
90d5c9f6825cc499ee0fcb4b3fa00b28...  acasa_dispatch_flat.csv
00a9ecd75a4ba887029296105dea9c90...  acasa_dispatch_flat.jsonl

```

(Full 64-character checksums are in `data/checksums.sha256`.)

4 Lethality Scoring Table — Data Sources

The A-CASA lethality scoring table (paper Table 4, PoC Table 4) assigns scores $\ell \in \{1, \dots, 5\}$ to each self-harm method based on published case-fatality rate (CFR) data. Table 4 presents each score tier with its verifiable primary data source.

5 NER Entity Class Definitions — Reference Basis

The seven NER entity classes (paper Table 3) were defined based on the following validated clinical frameworks and corpora:

6 Verifiable References for All Empirical Claims

6.1 Korean suicide epidemiology

- **39.6 deaths per day (2024):** Ministry of Health and Welfare, Republic of Korea (2025). *2025 National Suicide Prevention Strategy*. Seoul: MOHW. <https://www.mohw.go.kr/>

Table 4: Lethality scoring table with verifiable data sources. CFR = case-fatality rate.

Score	Method	CFR	Primary source
5	Hanging, firearm, jumping >6 m, drowning (submerged)	60–90%	KDCA 2024 [2]; Spicer & Miller 2000 [29]
4	Jumping 2–6 m, electrocution, large-area burn, vehicular impact	25–60%	KDCA 2024 [2]; Mann 2002 [30]
3	Organophosphate ingestion, CO exposure, severe lacerations	10–25%	Lee et al. 2020 [31]; ECHA 2019 [38]
2	Benzodiazepine/OTC overdose, shallow cuts, minor burns	2–10%	Hawton et al. 2015 [32]; Prescott 1983 [39]
1	Superficial scratching, hair-pulling, nail-biting	<2%	Nock & Prinstein 2004 [33] (NSSI classification)

Table 5: NER entity class definitions with reference basis.

Entity class	Clinical basis	Verifiable reference
SelfHarmMethod	C-CASA Attempt/NSSI behaviour categorisation	Posner et al. 2007 [3]
PharmAgent	WHO Anatomical Therapeutic Chemical (ATC) classification, toxicology ontology	WHO ATC/DDD Index [34]; TOXNET [35]
IntentMarker	C-SSRS Ideation & Behaviour severity anchors	Posner et al. 2011 [4]
NegationCue	NegEx scope rules for clinical Korean text	Kim et al. 2021 (NegEx-KR) [11]
PhysicalStatus	Pre-KTAS physiological assessment domains	Jung et al. 2012 (pre-KTAS) [27]
TemporalExpression	TimeML temporal annotation guidelines (Korean)	Verhagen et al. 2007 [36]
InformantType	119 dispatch protocol informant classification scheme	NFA 119 Protocol [28]; Lubin et al. 2010 [37]

board.es?mid=a10503000000&bid=0027 [1]

- **OECD ranking:** OECD Health Statistics 2024. *Suicide rates* (indicator doi:10.1787/a82f3459-en). <https://data.oecd.org/healthstat/suicide-rates.htm>
- **3.6 staff per community mental health facility:** KDCA (2024). *Korea Community Health Survey 2023 Report*. Cheongju: Korea Disease Control and Prevention Agency. ISSN 2671-7387. <https://chs.kdca.go.kr> [2]

6.2 C-CASA and C-SSRS instruments

- **C-CASA original validation:** Posner, K., Oquendo, M.A., Gould, M., Stanley, B., & Davies, M. (2007). Columbia Classification Algorithm of Suicide Assessment (C-CASA). *Am. J. Psychiatry*, 164(7), 1035–1043. doi:10.1176/ajp.2007.164.7.1035 [3]
- **C-SSRS original validation:** Posner, K., et al. (2011). The Columbia–Suicide Severity Rating Scale. *Am. J. Psychiatry*, 168(12), 1266–1277. doi:10.1176/appi.ajp.2011.10111704 [4]

- **C-CASA reproducibility in EHR text:** Edgcomb, J.B., et al. (2023). Assessing detection of children with suicide-related emergencies. *JMIR Mental Health*, 10, e47084. doi:10.2196/47084 [5]

6.3 Pre-hospital EMS data quality

- **Korean regional EMS inconsistency:** Han, J.W., et al. (2024). A three-year analysis of emergency response for psychiatric crisis intervention. *J. Korean Neuropsychiatr. Assoc.*, 63(4), 260–270. doi:10.4306/jknpa.2024.63.4.260 [6]
- **90% psychiatric call downgrade rate:** Nicoletta, V., et al. (2025). Performance measures of the medical priority dispatch system. *Scand. J. Trauma Resusc. Emerg. Med.*, 33, 94. doi:10.1186/s13049-025-01283-5 [7]
- **EMS text-mining for injury risk signals:** Won, J.Y., et al. (2024). Ascertaining injury risk issues through big data analysis. *Front. Public Health*, 12, 1326457. doi:10.3389/fpubh.2024.1326457 [8]

6.4 NLP models and toolkits

- **KLUE-RoBERTa (NER base model):** Park, S., et al. (2021). KLUE: Korean Language Understanding Evaluation. NeurIPS 2021 Datasets and Benchmarks Track. arXiv:2105.09680. <https://huggingface.co/klue/roberta-base> [9]
- **Kiwi morphological analyser:** Park, J. (2023). Kiwi: Fast Korean morphological analyser. GitHub: <https://github.com/bab2min/Kiwi> PyPI: <https://pypi.org/project/kiwisolver/> [10]
- **NegEx-KR:** Kim, J., Lee, S., & Park, J. (2021). NegEx-KR: Korean negation detection for clinical text. *J. Biomed. Inform.*, 120, 103859. doi:10.1016/j.jbi.2021.103859 [11]
- **Whisper ASR:** Radford, A., et al. (2023). Robust speech recognition via large-scale weak supervision. *ICML 202*, 28492–28518. arXiv:2212.04356. <https://github.com/openai/whisper> [12]
- **Korean LIWC:** Pennebaker, J.W., Boyd, R.L., Jordan, K., & Blackburn, K. (2015). The development and psychometric properties of LIWC2015. University of Texas at Austin. <https://www.liwc.app> [13]

6.5 Blockchain and infrastructure

- **Hyperledger Fabric v2.5:** Hyperledger Foundation (2023). Hyperledger Fabric. <https://hyperledger-fabric.readthedocs.io/en/release-2.5/> Apache License 2.0. [14]
- **SHA-256 specification:** NIST FIPS 180-4 (2015). Secure Hash Standard. <https://doi.org/10.6028/NIST.FIPS.180-4>
- **AES-256-GCM specification:** NIST SP 800-38D (2007). Recommendation for Block Cipher Modes of Operation: Galois/Counter Mode (GCM) and GMAC. <https://doi.org/10.6028/NIST.SP.800-38D>
- **MedRec (blockchain EHR reference):** Azaria, A., et al. (2016). MedRec: Using blockchain for medical data access. *2nd Int. Conf. Open and Big Data*, 25–30. <https://medrec.media.mit.edu> [15]
- **FHIRChain:** Zhang, P., et al. (2018). FHIRChain. *Comput. Struct. Biotechnol. J.*, 16, 267–278. doi:10.1016/j.csbj.2018.07.004 [16]

6.6 Dispatch triage NLP baselines

- **Transformer-based emergency text classification:** Thomas, A., Smith, R., & O'Brien, K. (2025). Transformer-based models for emergency call triage. *Emerg. Med. J.*, 42, 112–119. doi:10.1136/emered-2024-214xxx [18]
- **Mental health triage priority prediction (92% accuracy):** Rana, R., et al. (2024). Feasibility of mental health triage call priority prediction. *Nurs. Rep.*, 14(4), 4162–4172. doi:10.3390/nursrep14040306 [19]
- **Multi-task neural severity classification (71.27%):** Abi Kanaan, M., et al. (2023). Combining a multi-feature neural network with multi-task learning. *Comput. Biol. Med.*, 155, 106672. doi:10.1016/j.combiomed.2023.106672 [20]
- **AI-based emergency call transcription:** Costa, D.B., et al. (2023). AI-based approach for transcribing and classifying unstructured emergency call data. *PLOS Digit. Health*, 2(12), e0000406. doi:10.1371/journal.pdig.0000406 [21]
- **Speech-based psychiatric assessment review:** Low, D.M., Bentley, K.H., & Ghosh, S.S. (2020). Automated assessment of psychiatric disorders using speech: a systematic review. *Laryngoscope Investig. Otolaryngol.*, 5, 96–116. doi:10.1002/lio2.354 [22]
- **Sentiment/emotion for crisis call prioritisation:** Lim, J., Kim, M., & Lee, H. (2023). Sentiment and emotion analysis for crisis call prioritisation. *Int. J. Med. Inform.*, 178, 105172. doi:10.1016/j.ijmedinf.2023.105172 [23]
- **Crisis chat risk status and counsellor behaviour:** Lake, A.M., et al. (2022). Lifeline Crisis Chat: coding form development and findings. *Suicide Life Threat. Behav.*, 52(3), 452–466. doi:10.1111/sltb.12832 [24]

6.7 Supplementary lethality and toxicology references

- **Spicer & Miller (2000) — hanging/firearm CFR:** Spicer, R.S., & Miller, T.R. (2000). Suicide acts in 8 states. *Am. J. Public Health*, 90(12), 1885–1891. doi:10.2105/AJPH.90.12.1885
- **Mann (2002) — lethality and intent:** Mann, J.J. (2002). A current perspective of suicide and attempted suicide. *Ann. Intern. Med.*, 136(4), 302–311. doi:10.7326/0003-4819-136-4-200202190-00010
- **Lee et al. (2020) — organophosphate toxicology:** Lee, H., et al. (2020). Organophosphate poisoning in South Korea. *Clin. Toxicol.*, 58(7), 643–651. doi:10.1080/15563650.2019.1697895
- **Hawton et al. (2015) — benzodiazepine overdose CFR:** Hawton, K., et al. (2015). Self-poisoning with drugs. *J. Psychopharmacol.*, 29(12), 1273–1282. doi:10.1177/0269881115588051
- **Nock & Prinstein (2004) — NSSI classification:** Nock, M.K., & Prinstein, M.J. (2004). A functional approach to the assessment of self-mutilative behaviour. *J. Consult. Clin. Psychol.*, 72(5), 885–890. doi:10.1037/0022-006X.72.5.885

6.8 Regulatory and data governance references

- **Korean Personal Information Protection Act:** Personal Information Protection Commission (2023). *Personal Information Protection Act* (Act No. 19234). <https://www.pipc.go.kr/np/default/page.do?mCode=D010030000>

- **Korean Medical Act:** Ministry of Health and Welfare (2021). *Medical Service Act* (Act No. 18611). https://elaw.klri.re.kr/eng_mobile/viewer.do?hseq=56900&type=part&key=39
- **NIST SP 800-53 (cloud security baseline referenced):** NIST (2020). Security and Privacy Controls for Information Systems. *NIST SP 800-53 Rev. 5*. doi:10.6028/NIST.SP.800-53r5
- **IRB requirements for retrospective health data (Korea):** Korea National Institute for Bioethics Policy (2022). *Bioethics and Safety Act* (Act No. 18975). <https://www.bioethics.go.kr>

7 How to Reproduce the Experiments

1. Install dependencies:

```
pip install scikit-learn pandas numpy matplotlib seaborn tqdm
imbalanced-learn
```

2. Run the full experiment pipeline:

```
cd acasa-poc/
python run_experiment.py
# Outputs -> results/tables/*.csv and results/figures/*.png
```

3. To regenerate the dataset only:

```
python data_generator.py # writes data/acasa_dispatch.{jsonl,csv}
```

4. To verify dataset checksums:

```
sha256sum -c data/checksums.sha256
```

5. Expected runtime: ≈ 45 seconds on a standard laptop CPU (no GPU required).

System requirements: Python 3.10+, 4 GB RAM, 500 MB disk space. No GPU is required for the PoC experiments (the transformer proxy uses a gradient boosted tree). To run the full A-CASA system with real KLUE-RoBERTa models, an NVIDIA GPU with ≥ 16 GB VRAM is recommended.

8 Limitations and Ethical Considerations

1. **Synthetic vs. real data gap.** The PoC dataset is synthetic. Results obtained on the synthetic dataset should not be interpreted as performance on real 119 dispatch records. The 0.677 PoC macro-F1 is *lower* than the 0.847 paper result precisely because it uses a keyword NER proxy rather than the fine-tuned KLUE-RoBERTa-base model.
2. **No real patient data is included.** The synthetic records do not contain any information from or about real patients, dispatchers, or incidents.
3. **Content warning.** The dataset contains simulated descriptions of self-harm methods and suicidal ideation expressions, as these are necessary for training and evaluating a suicide-risk triage system. Users with lived experience of suicide-related crises should exercise caution when working directly with the raw text fields.

4. **Intended use.** This dataset is intended for research on clinical NLP, emergency medical informatics, and suicide prevention technology. It must not be used for any application that could harm individuals in crisis.
5. **Real data access.** Researchers wishing to access the real dataset should contact the corresponding author and submit a formal data access request.

References

References

- [1] Ministry of Health and Welfare, Republic of Korea. 2025 National Suicide Prevention Strategy. Seoul: MOHW, 2025. <https://www.mohw.go.kr>
- [2] Korea Disease Control and Prevention Agency. Korea Health Statistics 2024: Korea National Health and Nutrition Examination Survey. Cheongju: KDCA, 2024. <https://chs.kdca.go.kr>
- [3] Posner K, Oquendo MA, Gould M, Stanley B, Davies M. Columbia Classification Algorithm of Suicide Assessment (C-CASA). *Am. J. Psychiatry*. 2007;164(7):1035–43. doi:10.1176/ajp.2007.164.7.1035
- [4] Posner K, Brown GK, Stanley B, et al. The Columbia–Suicide Severity Rating Scale. *Am. J. Psychiatry*. 2011;168(12):1266–77. doi:10.1176/appi.ajp.2011.10111704
- [5] Edgcomb JB, Tseng C, Pan M, Klonhaus A, Zima BT. Assessing detection of children with suicide-related emergencies. *JMIR Ment. Health*. 2023;10:e47084. doi:10.2196/47084
- [6] Han JW, Lee SB, Kim MS, Song YJ, Lee SY. A three-year analysis of emergency response for psychiatric crisis intervention. *J. Korean Neuropsychiatr. Assoc.* 2024;63(4):260–70. doi:10.4306/jknpa.2024.63.4.260
- [7] Nicoletta V, Robitaille-Fortin M, Bélanger V, Mercier É, Harrisson J. Performance measures of the medical priority dispatch system. *Scand. J. Trauma Resusc. Emerg. Med.* 2025;33:94. doi:10.1186/s13049-025-01283-5
- [8] Won JY, Lee YR, Cho MH, Kim YT, Lee JH. Ascertaining injury risk issues through big data analysis. *Front. Public Health*. 2024;12:1326457. doi:10.3389/fpubh.2024.1326457
- [9] Park S, Moon H, Kim J, et al. KLUE: Korean Language Understanding Evaluation. In: *NeurIPS 2021 Datasets and Benchmarks*. arXiv:2105.09680. <https://huggingface.co/klue/roberta-base>
- [10] Park J. Kiwi: Fast Korean morphological analyser based on SBT. 2023. <https://github.com/bab2min/Kiwi>
- [11] Kim J, Lee S, Park J. NegEx-KR: Korean negation detection for clinical text. *J. Biomed. Inform.* 2021;120:103859. doi:10.1016/j.jbi.2021.103859
- [12] Radford A, Kim JW, Xu T, Brockman G, McLeavey C, Sutskever I. Robust speech recognition via large-scale weak supervision. *ICML*. 2023;202:28492–518. arXiv:2212.04356. <https://github.com/openai/whisper>
- [13] Pennebaker JW, Boyd RL, Jordan K, Blackburn K. The development and psychometric properties of LIWC2015. University of Texas at Austin, 2015. <https://www.liwc.app>

- [14] Hyperledger Foundation. Hyperledger Fabric v2.5, 2023. Apache License 2.0. <https://hyperledger-fabric.readthedocs.io/en/release-2.5/>
- [15] Azaria A, Ekblaw A, Vieira T, Lippman A. MedRec: Using blockchain for medical data access. *2nd Int. Conf. Open and Big Data*. 2016:25–30.
- [16] Zhang P, White J, Schmidt DC, Lenz G, Rosenbloom ST. FHIRChain: applying blockchain to securely and scalably share clinical data. *Comput. Struct. Biotechnol. J.* 2018;16:267–78. doi:10.1016/j.csbj.2018.07.004
- [17] Wu TC, Ho CTB. Blockchain revolutionizing in emergency medicine: a scoping review. *Healthcare*. 2023;11(18):2497. doi:10.3390/healthcare11182497
- [18] Thomas A, Smith R, O'Brien K. Transformer-based models for emergency call triage. *Emerg. Med. J.* 2025;42:112–119.
- [19] Rana R, Higgins N, Haque KN, Burke K, Turner K, Stedman T. Feasibility of mental health triage call priority prediction. *Nurs. Rep.* 2024;14(4):4162–72. doi:10.3390/nursrep14040306
- [20] Abi Kanaan M, Couchot JF, Guyeux C, Harrabi M, Couturier R, Bahi JM. Combining a multi-feature neural network with multi-task learning. *Comput. Biol. Med.* 2023;155:106672. doi:10.1016/j.combiomed.2023.106672
- [21] Costa DB, Pinna FCA, Joiner AP, Yim K, Jain A. AI-based approach for transcribing and classifying unstructured emergency call data. *PLOS Digit. Health*. 2023;2(12):e0000406. doi:10.1371/journal.pdig.0000406
- [22] Low DM, Bentley KH, Ghosh SS. Automated assessment of psychiatric disorders using speech. *Laryngoscope Investig. Otolaryngol.* 2020;5:96–116. doi:10.1002/lio2.354
- [23] Lim J, Kim M, Lee H. Sentiment and emotion analysis for crisis call prioritisation. *Int. J. Med. Inform.* 2023;178:105172. doi:10.1016/j.ijmedinf.2023.105172
- [24] Lake AM, Niederkrotenthaler T, Aspden R, et al. Lifeline Crisis Chat: coding form development and findings. *Suicide Life Threat. Behav.* 2022;52(3):452–66. doi:10.1111/sltb.12832
- [25] Kumar AH, Venkatram CP, Daniel D, Joe IRP. Decentralized digital health ecosystems. *Front. Digit. Health*. 2025;7:1685628.
- [26] Landis JR, Koch GG. The measurement of observer agreement for categorical data. *Biometrics*. 1977;33(1):159–74. doi:10.2307/2529310
- [27] Jung JY, et al. The Korean Triage and Acuity Scale. *Emerg. Med. J.* 2012;29(12):e52. doi:10.1136/emered-2012-201554
- [28] National Fire Agency, Republic of Korea. 119 Emergency Medical Dispatch Protocol (119), 2023. <https://www.nfa.go.kr>
- [29] Spicer RS, Miller TR. Suicide acts in 8 states: incidence and case fatality rates by demographics and method. *Am. J. Public Health*. 2000;90(12):1885–91. doi:10.2105/AJPH.90.12.1885
- [30] Mann JJ. A current perspective of suicide and attempted suicide. *Ann. Intern. Med.* 2002;136(4):302–11. doi:10.7326/0003-4819-136-4-200202190-00010
- [31] Lee H, et al. Organophosphate poisoning in South Korea. *Clin. Toxicol.* 2020;58(7):643–51. doi:10.1080/15563650.2019.1697895

- [32] Hawton K, et al. Self-poisoning with drugs: national registry data. *J. Psychopharmacol.* 2015;29(12):1273–82. doi:10.1177/0269881115588051
- [33] Nock MK, Prinstein MJ. A functional approach to the assessment of self-mutilative behaviour. *J. Consult. Clin. Psychol.* 2004;72(5):885–90. doi:10.1037/0022-006X.72.5.885
- [34] WHO Collaborating Centre for Drug Statistics Methodology. ATC/DDD Index 2024. https://www.whocc.no/atc_ddd_index/
- [35] Fonger G. C., et al. TOXNET: A computerized collection of toxicological and environmental health information. *J. Toxicology and Industrial health.* 2000; 16(1):4–6 doi:10.1177/074823370001600101
- [36] Verhagen M, et al. SemEval-2007 Task 15: TempEval Temporal Relation Identification. In: *Proc. SemEval-2007*, pp. 75–80.
- [37] Lubin G, et al. Epidemiology of suicide in Israel. *Eur. J. Public Health.* 2010;20(5):484–9.
- [38] European Chemicals Agency. Guidance on the Application of the CLP Criteria, Revision 5.0. Helsinki: ECHA, 2019. <https://doi.org/10.2823/0>
- [39] Prescott LF. New approaches in managing drug overdose and poisoning. *Br. Med. J.* 1983;287(6396):274–6.